



A.2 PROGRAM COMPILATION

MIKROELEKTRONIKA

The microcontroller does not understand higher programming languages as such, so that it is necessary to compile the program into machine language. A single click on the appropriate icon within the compiler results in creating a new document with .hex. It is actually the same program, only compiled into machine language which the microcontroller perfectly understands. Such a program is usually called a *hex code* and forms a seemingly meaningless sequence of hexadecimal numbers. When compiled, the program should be loaded into the chip. And you need appropriate hardware to make this happen. In other words, you need a programmer.

```
:03000000020100FA1001000075813F
7590FFB29012010D80F97A1479D40
90110003278589EAF3698E8EB25B
A585FEA2569AD96E6D8FED9FAD
AF6DD00000001FF255AFED589EA
F3698E8EB25BA585FEA2569AD96
DAC59700D00000278E6D8FED9FA
DAF6DD00000001FF255AFED8FED
9FADAF6DD000F7590FFB29013278
E6D8FED9FADAF6DD00000001FF2
55AFED589EAF3698E8EB25BA585
FEA2569AD96DAC59D9FADAF6D
D00000001FF255AFED8FED9FADA
F6DD000F7590FFB29013278E6D82
78E6D8FED9FA589EAF3698E8EB2
5BA585FEA2569AD96DAF6DD000
00001FF2DAF6DD00000001FF255A
ADAF6DD00000001FF255AFED8FE
D9FA
```

As mentioned, to enable a *hex code* to be loaded into the microcontroller it is necessary to provide a special device, called the programmer, with appropriate software. A large number of programs and electronic circuits used for this purpose can be found on the Internet. The procedure is basically the same for all of them and looks as follows:

- Place the microcontroller in appropriate programmer's socket;
- Use an adequate cable to connect the programmer to a PC;
- Open hex code document within the programmer's software, set a few parameters and click the icon to transfer the code. After a few seconds, a sequence of zeros and ones is to be programmed into the microcontroller.

All that's left is to install the programmed chip into the target device. If it is necessary to make some changes in

the program, the previous procedure may be repeated an unlimited number of times.



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